AYMANE AIT DADS

Embedded AI Intern | AI Model Optimization · Embedded Systems · Hardware-Aware Deployment

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, $106K+ prize pool), demonstrating end-to-end AI model training, evaluation, and optimization on a 30B-parameter architecture under strict inference constraints. Reduced manual network analysis time by 60% at Orange Maroc by deploying Random Forest and K-Means pipelines on 100K+ records, translating model outputs directly into production maintenance recommendations. Brings hands-on experience in Python, PyTorch, ML model evaluation, ablation studies, and hardware-aware training (BF16, FP16 mixed precision, gradient accumulation) directly applicable to embedded AI performance benchmarking and trade-off analysis for FPGA and MCU platforms. Aims to contribute to AUMOVIO's automotive embedded AI roadmap by evaluating AI/ML models under resource constraints and proposing optimization strategies for latency, memory, and accuracy.

CORE COMPETENCIES

AI Model Evaluation | Embedded AI Optimization | FPGA-Based AI Acceleration | Resource-Constrained Deployment | Performance Benchmarking | Hardware-Aware AI | ML Model Optimization | Accuracy-Latency Trade-off Analysis

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built and deployed Random Forest and K-Means ML models on **100K+ fiber-optic network records**, evaluating performance across upload speed, latency, and jitter metrics for network quality classification.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **60%** through automated feature engineering and model evaluation scripts, accelerating actionable insight delivery to stakeholders.
- Delivered technical presentations translating model outputs into **maintenance recommendations**, documenting trade-offs between classification approaches using Power BI dashboards and Python reports.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge — *Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams*

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified CoT data.

PyTorch · Unsloth · PEFT · vLLM · Triton · Kaggle GPU

AI-Generated Image Detection — *EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class*

- Ranked **1st in the EURECOM class private Kaggle leaderboard** (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M params) with a custom MLP head, dual learning rates, and FP16 mixed-precision training on 250K images across 25+ generator types.
- Evaluated accuracy-efficiency trade-offs across 1-epoch vs. 4-epoch training regimes - finding that **1-epoch generalization (0.793 AUC) outperformed 4-epoch (0.767 AUC)** on out-of-distribution data; submitted results to the NTIRE 2026 @ CVPR CodaBench international benchmark.

PyTorch · OpenCLIP · AdamW · Kaggle GPU · FP16

Anomalous Sound Detection - Industrial Equipment — *EURECOM · 2025 · Unsupervised Fault Detection*

- Designed an unsupervised Transformer autoencoder on Mel spectrograms with SpecAugment for fault detection on slide-rail machinery, achieving **AUC 0.80** for predictive maintenance without labeled anomaly data.
- Benchmarked model performance under constrained training conditions, analyzing **accuracy-efficiency trade-offs** between reconstruction error thresholds and false-positive rates across operational frequency bands.

PyTorch · Mel Spectrogram · SpecAugment · Transformer Autoencoder

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM — *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Image Security, Cloud Computing, Statistics

CPGE - Mathematics & Physics - Ibn Timiya — *2021 - 2023*

SKILLS

ML Frameworks & Model Optimization: PyTorch | Scikit-learn | Hugging Face Transformers | PEFT | Unsloth | TRL | vLLM

AI/ML Techniques: LoRA SFT | BF16/FP16 Mixed Precision | Ablation Studies | Ensemble Methods | Computer Vision | Feature Engineering

Programming & Tools: Python | SQL | Bash | Git | Linux | Power BI | Docker (basic)